Genetic variation through protein interactions

Genetic landscape of an in vivo protein interactome

Besse, Sakaguchi, Gauthier et al. — Nature Genetics 58, 2684–2695 (2026)

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The question: where do distant genetic effects go?

A variant can change a protein interaction without changing either partner's coding sequence.

1. It may alter transcription, translation, degradation or localization.
2. It may change a competing binder, modification or cellular environment.
3. These effects converge on the amount of a protein complex in living cells.

Central hypothesis
In vivo protein interactions can serve as molecular reporters linking natural genetic variation to cellular responses.

Journal abstract and Fig. 1a; preprint Introduction.

What this study measures — and what it does not

Input
354 haploid yeast strains descended from wine and clinical isolates; 12,054 segregating SNPs.

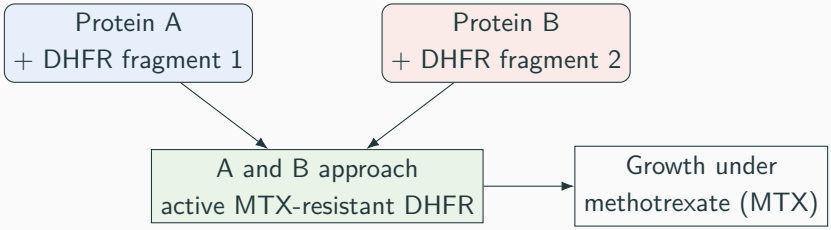
Readout
61 protein-pair reporters involving 44 proteins, assayed across five environments.

Output
Protein-interaction quantitative trait loci: piQTLs, variants associated with an interaction reporter.

Scope
A selected subnetwork enriched around ergosterol synthesis, rather than an exhaustive interactome census.

Journal Fig. 1b,d and Extended Data Fig. 1; preprint Results and Methods.

Make a protein interaction selectable

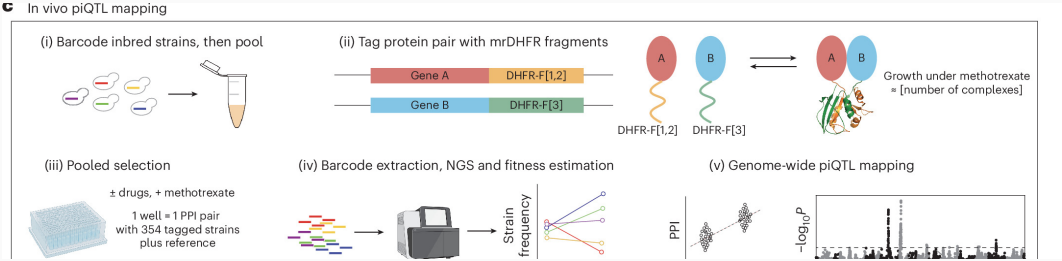


The two endogenous protein genes receive complementary C-terminal tags. Proximity can reconstitute the reporter enzyme.

Interpretation: growth reports effective interaction/complex formation, integrating abundance and cellular context; it is not an isolated binding-affinity measurement.

Journal Fig. 1c and Extended Data Fig. 2; preprint Results, pp. 5–8. Diagram adapted.

Pool strains, then decode their growth by sequencing



One pool contains different genetic backgrounds carrying the *same* protein-pair reporter. Separate pools assay the other pairs.

Neutral chromosomal barcodes identify strains; untagged spike-ins provide a reference without DHFR complementation.

Journal Fig. 1c; Supplementary Methods 2.2–2.3.

A running example: from barcode counts to a piQTL

Let $f_{i,t}$ be the barcode frequency of strain i at time t.

$$y_i = \log_2 \frac{f_{i,96h}}{f_{i,0h}} - \text{mean}_{r \in \text{untagged}} \left[\log_2 \frac{f_{r,96h}}{f_{r,0h}} \right].$$

Illustrative numbers: a tagged strain doubles its frequency while the reference halves its frequency: $y_i = 1 - (-1) = 2$.

Repeat across strains, then test each SNP for association with y_i for that protein pair and environment.

A genotype-associated shift in this readout defines a candidate piQTL; the readout is relative to pooled competition and the reference strains.

Preprint Methods: Fitness and PPI estimation. Toy numbers are illustrative, not observations.

Five environments probe genetic context

Condition	Rationale
No added drug	Baseline; MTX still selects the interaction reporter
Fluconazole	Perturbs ergosterol synthesis; Erg11 is a direct target
5-fluorocytosine	Perturbs nucleotide metabolism and protein synthesis
Metformin	Broad metabolic perturbation; useful for network hypotheses
Trifluoperazine	Antipsychotic; yeast response probes conserved cellular processes

The same reporter can have different genetic determinants across environments. Human drug labels do not make yeast a direct model of clinical efficacy.

Journal Fig. 2; Supplementary Methods 2.3; preprint Results.

Controls are central because the assay is based on growth

1. **Replication:** independent biological replicates and doubly barcoded strains assess reproducibility.
2. **Reference:** two untagged strains estimate growth without reporter complementation.
3. **Masking:** remove loci associated with basal medium, tagging-specific growth and MTX response, including linked SNPs.

A substantial assay effect

47 of 61 pairs had tagging-sensitive growth QTLs in medium without MTX or environmental drug. These were masked separately for each pair.

Masking reduces confounding but also removes parts of the genome from discovery.

Journal Supplementary Methods 2.2d and 2.9a–b; Extended Data Fig. 3.

Map associations without counting linked SNPs repeatedly

For a pair and condition, regress the reporter across strains on SNP genotype, allowing for residual genetic structure.

Journal analysis: select lead SNPs in residual LD blocks at $r^2 = 0.50$, with additional signals surviving forward regression.

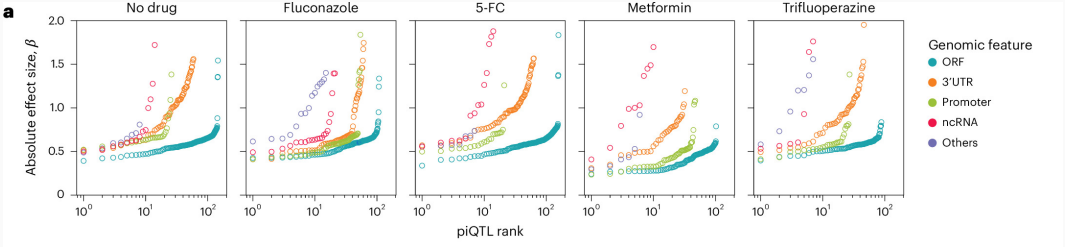
Association events	Unique SNPs	Unique gene loci
1,180	354	282

About four events per protein pair per condition. A shared locus can count in multiple pairs and environments.

Version note: the 2023 preprint used different LD/counting choices. Its headline counts are not used here.

Journal Extended Data Fig. 3d; Supplementary Methods 2.9.

Result 1: regulatory regions carry substantial effects

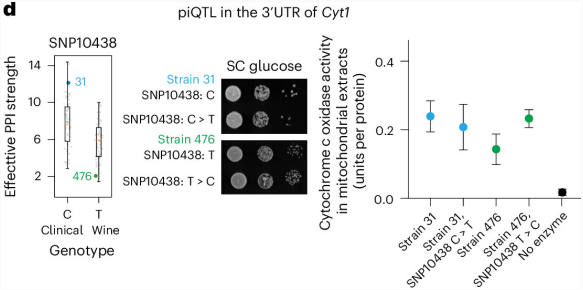


Across conditions, 3′UTR and ncRNA variants can have effects comparable to, or larger than, coding variants.

The horizontal axis is *rank within feature*, not genomic position. These are selected significant effects, rather than the distribution of all tested variants.

Journal Fig. 2a; effect size is the association coefficient in reporter units.

A 3′UTR example connects a reporter to respiration

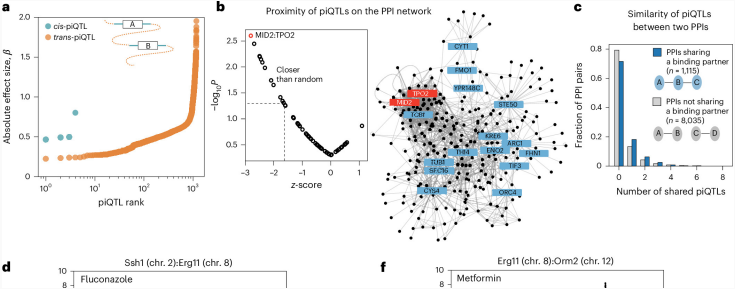


SNP10438 in the *CYT1* 3′UTR separates reporter distributions. Reciprocal allele edits were examined in strains 31 and 476.

The right panel measures cytochrome c oxidase activity in mitochondrial extracts; editing T to C raises activity in strain 476. The other background shows a smaller change.

Journal Fig. 2d; Supplementary Methods 2.5. Background dependence limits a universal allele-effect interpretation.

Result 2: genomic distance differs from network distance



Genomic view: only four cis events under a conservative 5-kb definition (three under fluconazole, one under metformin).

Network view: piQTL-encoded proteins can lie near one another; reporter pairs sharing a partner more often share piQTLs.

Journal Fig. 3a–c; Extended Data Fig. 6a–b.

How is network proximity tested?

For one reporter, collect the coding-region piQTL genes across conditions and map their proteins to the yeast PPI network.

1. Compute the mean shortest-path distance between these proteins.
2. Draw the same number of random network nodes 10,000 times.
3. Compare the observed distance with this null distribution using a *z*-score and bootstrap significance.

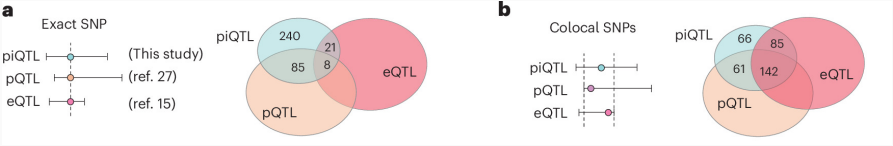
What this establishes

Genetically implicated proteins can occupy a coherent network neighborhood despite distant genomic locations.

Discussion: a uniform-node null may not fully account for degree or study-coverage differences; network proximity alone does not establish a causal route.

Journal Extended Data Fig. 8. Discussion is our interpretation.

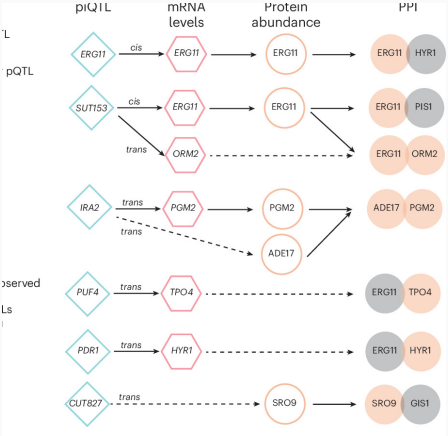
Result 3: piQTLs overlap protein QTLs more than eQTLs



Exact-SNP overlap: 93 piQTL SNPs overlap pQTLs (85 + 8), versus 29 with eQTLs (21 + 8). LD-aware overlap is also shown separately.

Journal Fig. 4a–b; Extended Data Fig. 9: size-matched permutation controls for reporter coverage.

Overlap supports several molecular paths

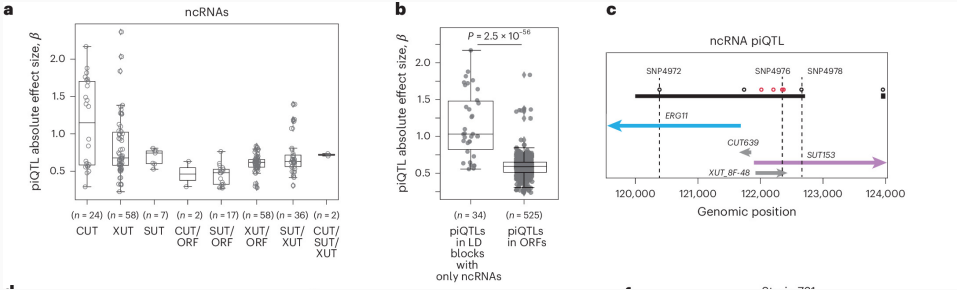


Some loci align with an mRNA-to-protein-to-complex route; others associate with protein abundance or PPIs without a detected mRNA effect.

Inference boundary: shared association or colocalization does not prove mediation. Absence of a detected eQTL also depends on power, conditions and coverage.

Journal Fig. 4e. Solid/dashed paths reproduce the authors' schematic; interpretation is qualified here.

Result 4: an ncRNA locus near the drug target

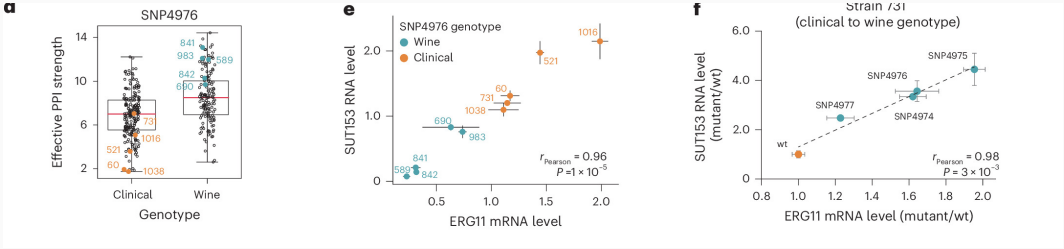


The *ERG11* neighborhood contains overlapping noncoding transcripts. Significant variants include SNP4976/4978 in *SUT153*.

Even LD blocks annotated only to ncRNAs show large selected effects (panel b), reducing a simple coding-variant explanation; it does not identify the responsible transcript by itself.

Journal Fig. 5a–c; Extended Data Fig. 6b. CUT, SUT and XUT denote transcript classes.

From correlation to allele replacement

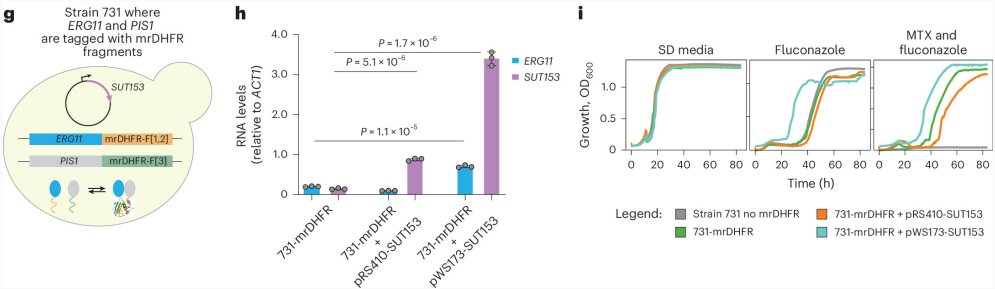


Panel d: genotype groups differ in the interaction reporter. Panel e: *SUT153* and *ERG11* RNA levels correlate across selected strains ($r = 0.96$).

Panel f: changing individual alleles from clinical to wine sequence in strain 731 changes both RNA levels. A shared background strengthens the inference beyond between-strain correlation.

Journal Fig. 5d–f; Supplementary Methods 2.4: two-step CRISPR/stuffer edits, verified by Sanger sequencing.

Does adding the ncRNA change the reporter phenotype?

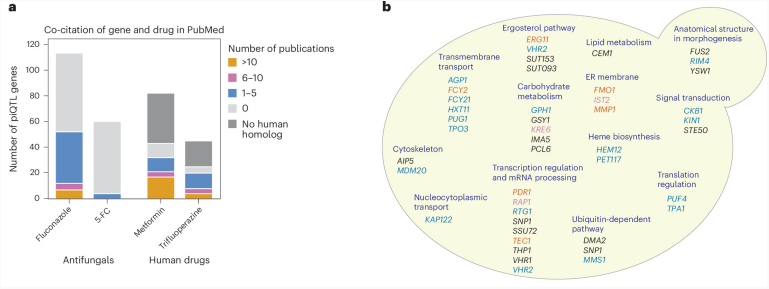


Plasmid-based *SUT153* expression is tested with tagged *ERG11:PIS1*. RNA levels and growth are compared in medium, fluconazole, and MTX + fluconazole.

The two constructs produce different expression/growth patterns. These perturbations support regulatory capacity, while leaving the endogenous mechanism and dose dependence open.

Journal Fig. 5g–i. Growth axis: OD_{600}; time: hours. No numeric kinetic model is inferred.

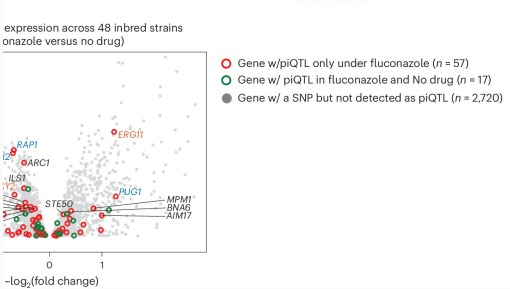
Result 5: drug-response loci form interpretable hypotheses



Fluconazole piQTLs implicate transport, lipid metabolism, stress responses and mitochondria. Known genes provide plausibility; poorly annotated genes become follow-up candidates.

Journal Fig. 6a–b; Supplementary Results 1.2: PET117 and RTG1 link respiration to resistance hypotheses.

An orthogonal response: transcription under fluconazole



RNA-seq compared 48 strains with and without fluconazole. The paired design models strain identity and environment; 4,278 genes have adjusted $P < 0.05$.

Fig. 6d highlights piQTL genes within that response. Drug-responsive expression and genetic control of a PPI are related questions, not interchangeable evidence.

Journal Fig. 6c–d; Supplementary Methods 2.8.

What survives a critical reading?

1. **Strong contribution:** a scalable experimental bridge from natural variants to in vivo interaction reporters, with tagging/MTX controls and targeted perturbations.
2. **Restricted generality:** two parental genomes, a selected 61-pair network and five environments; other proteins, variants and doses may differ.
3. **Mechanistic boundary:** reporter growth combines binding, abundance and localization. Network coherence and cross-omics overlap guide experiments rather than fully resolve causality.

Next experiment: measure protein abundance, localization and a second interaction assay in the same allele-edited strains and conditions.

Synthesis of journal figures, supplement and preprint Limitations; proposed experiment is our interpretation.

Discussion and source boundary

Discussion prompts

Which reporter pairs best separate abundance effects from interaction-specific regulation? Would degree-matched network nulls change the proximity result? How should variants lost to masking be revisited?

Sources used

Journal citation, abstract, all six original main figures and extended-data captions; 16-page journal supplement; full 52-page 2023 bioRxiv v1 including methods/supplement.

Access gap: journal main text/PDF remained behind subscription access; Chrome automation was unavailable. Current figure evidence is distinguished from preprint methods throughout. Revised count/LD/cis definitions follow journal sources.

DOIs: 10.1038/s41588-026-02747-z; 10.1101/2023.12.14.571726. See README for links and QA.